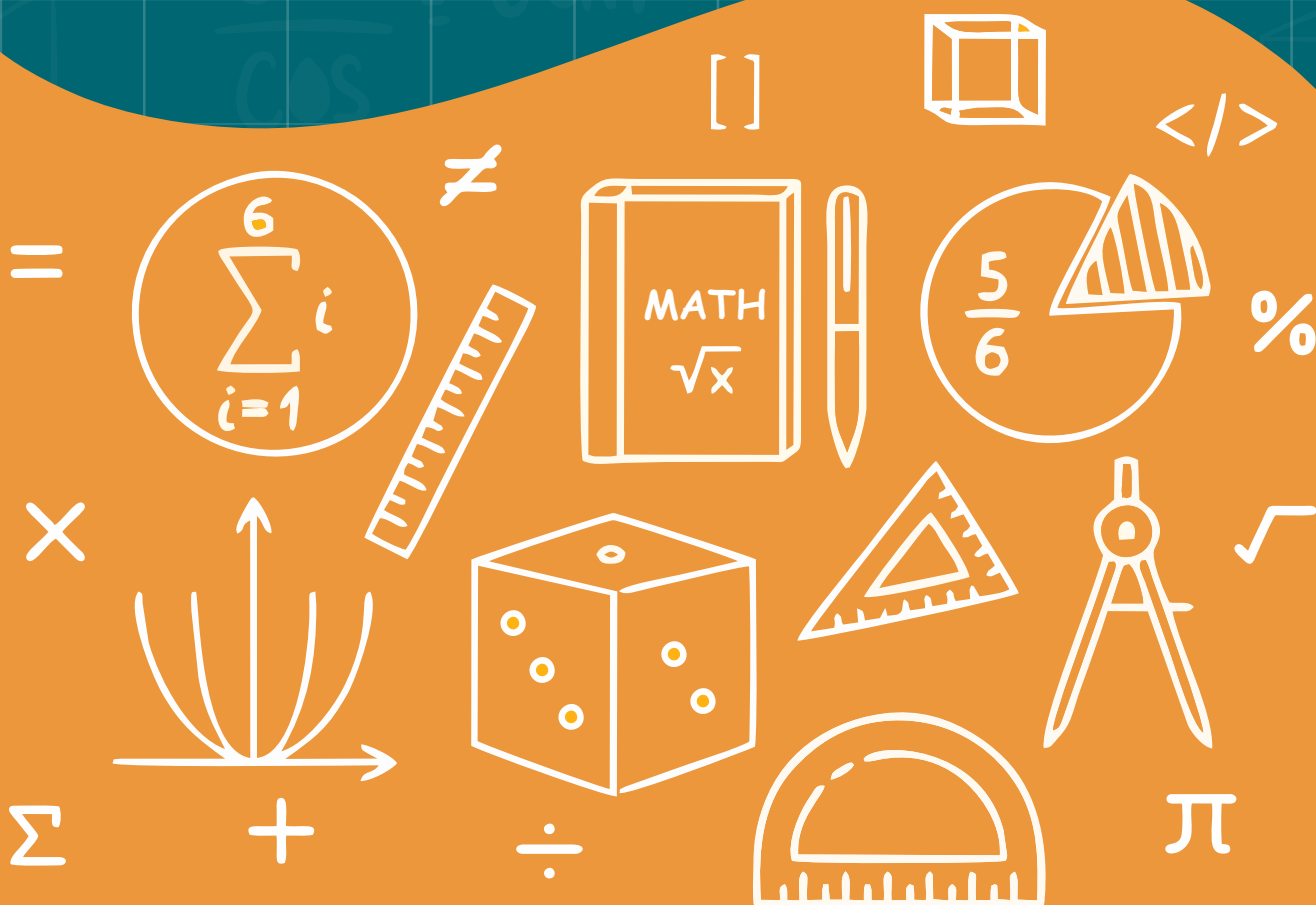


UNIT 4

INTEGERS



INTEGERS

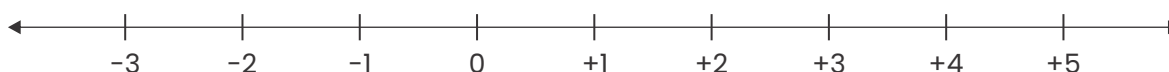
The natural numbers 1, 2, 3, ..., are also called positive integers and the corresponding negative numbers -1, -2, -3, ..., are called negative integers.

'0' is an integer which is neither positive nor negative.

ORDERING OF INTEGERS

Represent integers on number line.

We already know how to represent a natural number on a number line. Let us represent integers on a number line.



This line is called a number line.

On a number line, any number to left of zero is negative.

Any number lying to the right of zero is positive.

On a number line, any number to the left of another number is smaller.

On a number line, any number to the right of another number is greater.

Example: Decide whether first number is greater or smaller using number line.

(i) +5, -2

(ii) -6, 0

(iii) +4, 0

(iv) -3, -5

Solution:

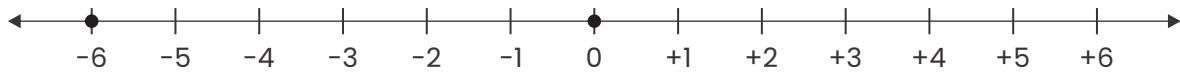
(i) Let us represent the numbers on number line.



The number +5 is at the right of -2

Therefore $+5 > -2$.

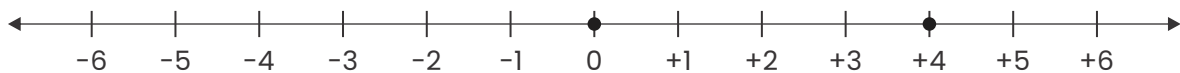
(ii) Let us represent the numbers on number line.



The number -6 is at the left of zero.

Therefore $-6 < 0$.

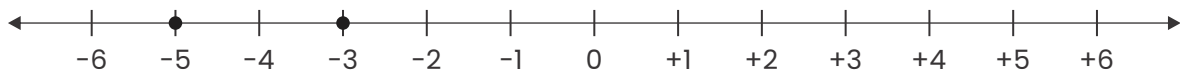
(iii) Let us represent the numbers on number line.



The number $+4$ is at the right of zero.

Therefore $+4 > 0$.

(iv) Let us represent the numbers on number line.



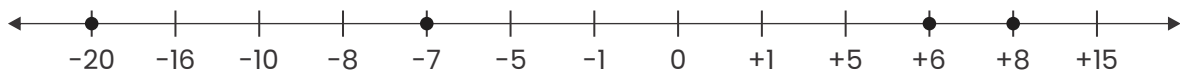
The number -3 is at the right of -5 .

Therefore $-3 > -5$.

Let us observe the relation between negative and positive numbers from the following examples:

(i) $+6 > -7$ **(ii)** $+1 > -20$ **(iii)** $+8 > -1$

Let us represent the number on number line.



It is observed that:

(i) $+6 > -7$ because $+6$ is on right of -7

(ii) $+1 > -20$ because $+1$ is on right of -20

(iii) $+8 > -1$ because $+8$ is on right of -1

From above examples it is clear that all positive numbers are on right of negative numbers.

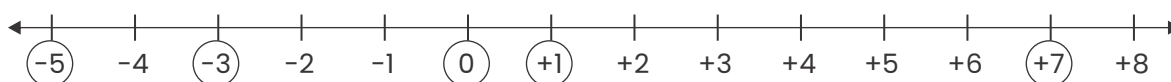
Every positive integer is greater than a negative integers.

Every negative integer is less than a positive integer.

Integers in ascending and descending order:

Example: Arrange the following numbers $-3, 0, 7, 1, -5$ in ascending and descending order.

Solution: First of all we indicate the given numbers ($-3, 0, 7, 1, -5$) on a number line as under.



Here the greatest number is 7 and the smallest number is -5 .

So, **Ascending order** (smallest to greatest) is:

$-5, -3, 0, 1, 7$

Descending order (greatest to smallest) is:

$7, 1, 0, -3, -5$

ABSOLUTE OR NUMERICAL VALUE OF AN INTEGER

Absolute value or numerical value of a number is its distance from zero on the number line and is always positive or zero.

Example: Write the absolute values of following numbers in ascending and descending orders. $+4, -2, +1, -3, 0, -5, +6$.

Solution:

Here,

$ +4 $	$=$	4	and	$ -3 $	$=$	3
$ -2 $	$=$	2		$ 0 $	$=$	0
$ +1 $	$=$	1		$ -5 $	$=$	5
				$ +6 $	$=$	6

Absolute values of the given integers are:

4, 2, 1, 3, 0, 5, 6

So, ascending order is:

0, 1, 2, 3, 4, 5, 6

And descending order is:

6, 5, 4, 3, 2, 1, 0

EXERCISE 4.1

Q1. Represent the following integers on a number line.

(i) -2, -1, 0, +1, +2

(ii) -3, -2, -1, 0, +1, +2, +3, +4

(iii) -4, -3, -2, -1, 0, +1, +2

(iv) +5, -5, -4, +3, +1, -2

Q2. Fill the blanks from given symbols.

(i) +10 _____ -20

($<$, $>$)

(ii) -16 _____ -4

($<$, $>$)

(iii) +25 _____ -100

($<$, $>$)

(iv) +30 _____ +50

($<$, $>$)

(v) -17 _____ +17

($<$, $>$)

(vi) 0 _____ -5

($<$, $>$)

Q3. Write True or False.

(i) Positive integer is always greater than negative integer. (____)

(ii) 0 is a positive integer. (____)

(iii) Every negative integer is less than zero. (____)

(iv) Absolute value of -5 is less than absolute of +4. (____)

(v) -5 is smaller than -10. (____)

(vi) -25 is greater than -100. (____)

(vii) Numerical value of a number can never be negative. (____)

Q4. Write the absolute values of the following.

(i) -5 (ii) +20 (iii) 0 (iv) -18 (v) +50

Q5. Arrange the given integers in ascending and descending order.

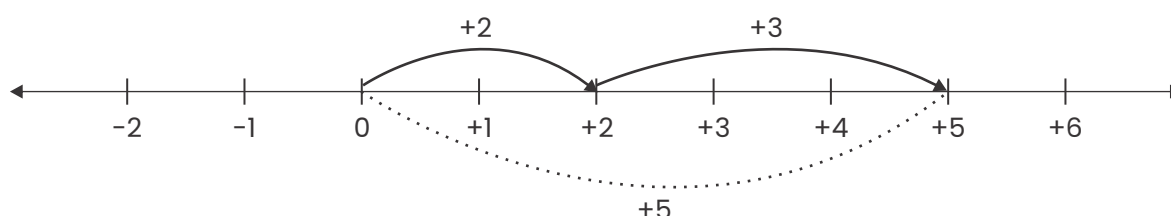
(i) +5, -7, +1, 0, -3, -1 (ii) -3, +4, 0, -1, +2, +5

(iii) 0, -4, +4, -5, +5 (iv) -4, -1, -7, -2, -8

ADDITION OF INTEGERS

Example 1: Add +2 and +3 using number line.

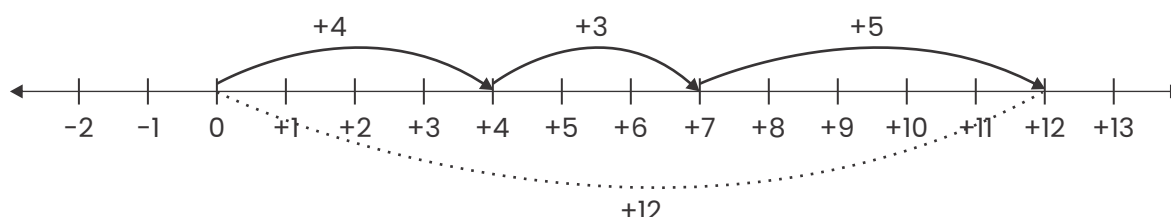
Solution:



Here, $(+2) + (+3) = +5$

Example 2: Find the sum: $(+4) + (+3) + (+5)$ using number line.

Solution:

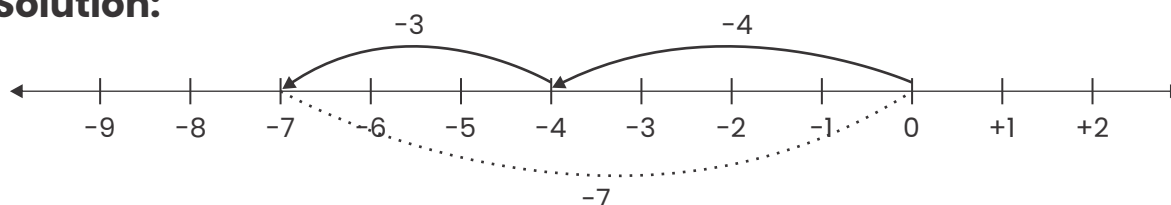


Here, $(+4) + (+3) + (+5) = (+12)$

Sum of two or more given negative integers

Example 1: Find the sum of -4 and -3.

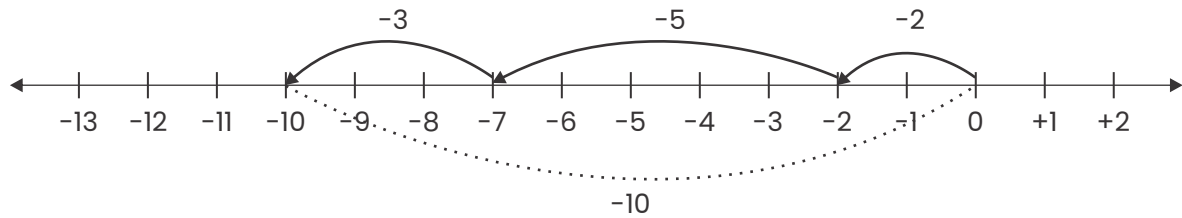
Solution:



Here, $(-4) + (-3) = (-7)$

Example 2: Find the sum: $(-2) + (-5) + (-3)$.

Solution:

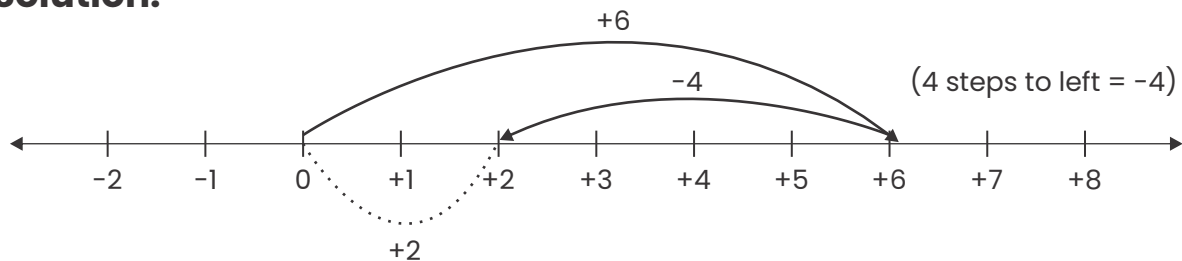


Here, $(-2) + (-5) + (-3) = (-10)$

Use number line to display difference of two given positive integers

Example 3: Add $+6$ and -4 .

Solution:



Here, $(+6) + (-4) = (+2)$

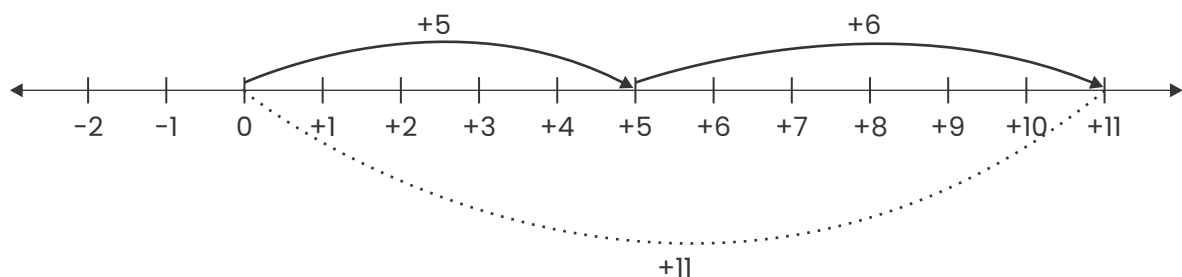
Use number line to display sum of two given integers:

Example 4: Add:

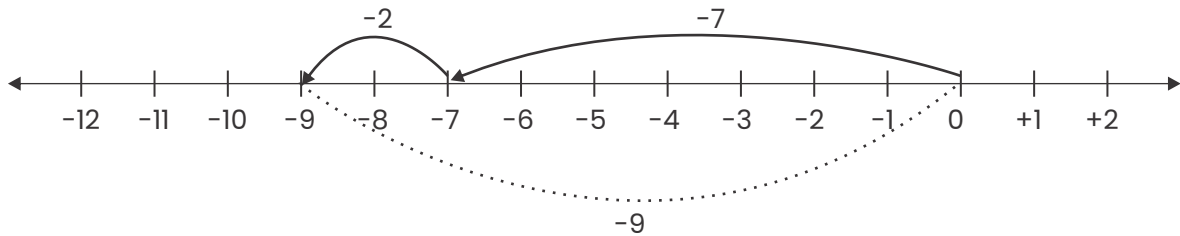
(i) $+5$ and $+6$ (ii) -7 and -2 (iii) $+8$ and -5 (iv) -6 and $+2$

Solution:

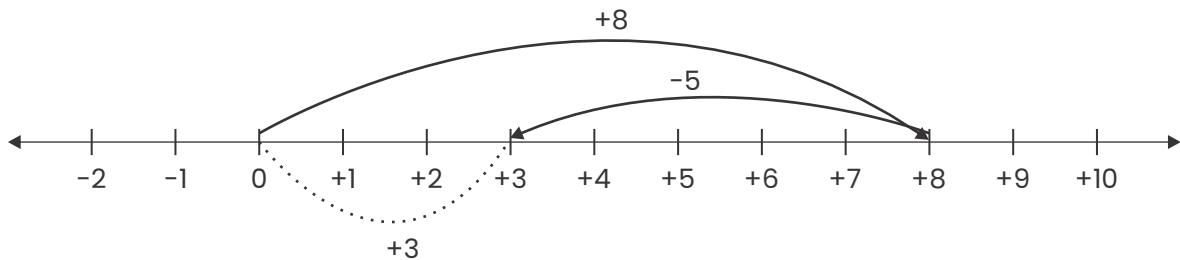
i) $+5$ and $+6$



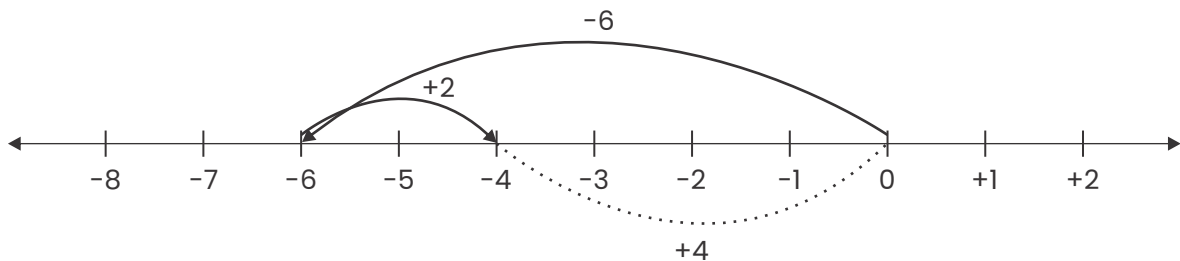
ii) Add -7 and -2 means to solve $(-7) + (-2)$



iii) +8 and -5



iv) -6 and +2



Addition of two integers (with like signs):

Example:

Solve:

(i) $(12) + (13)$

(ii) $(-10) + (-14)$

Solution:

(i) $(+12) + (+13)$

Here, $|+12| = 12$

$|+13| = 13$

Now $(+12) + (+13)$

$= + (12 + 13)$

$= + 25$

(ii) $(-10) + (-14)$

Here, $|-10| = 10$

$|-14| = 14$

Now $(-10) + (-14)$

$= - (10 + 14)$

$= - 24$

Example:**Solve:**

(i) $(15) + (-10)$

(ii) $(-20) + (12)$

Solution:

(i) $(+15) + (-10)$

Here, $|+15| = 15$

$| -10| = 10$

Now $(+15) + (-10)$

$= + (15 - 10)$

$= + 5$

(Sign of integer with greater absolute value is +)

(ii) $(-20) + (12)$

Here, $| -20| = 20$

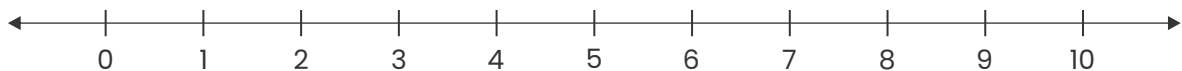
$| +12| = 12$

Now $(-20) + (+12)$

$= - (20 - 12)$

$= - 8$

(Sign of integer with greater absolute value is -)

EXERCISE 4.2**Q1. Find sum of the following using number line.**

$3 + 4 =$

$2 + 2 =$

$2 + 6 =$

$2 + 4 =$

$5 + 5 =$

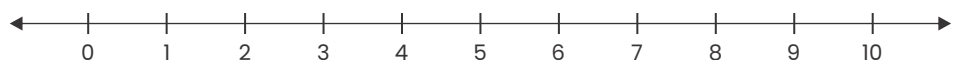
$5 + 2 =$

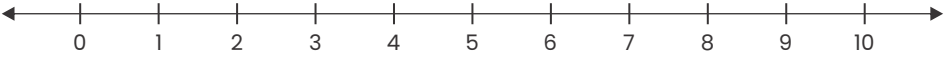
$2 + 7 =$

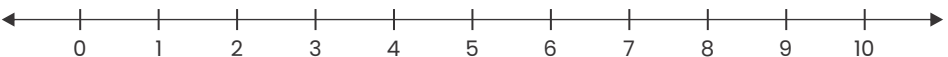
$7 + 3 =$

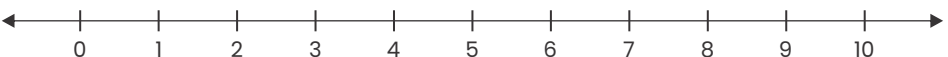
Q2. Solve the following.

$5 + 3 =$

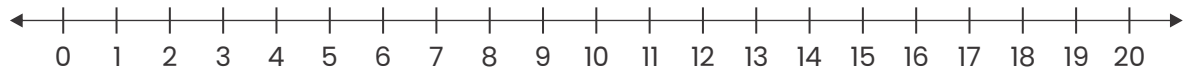


$1 + 2 = \boxed{}$


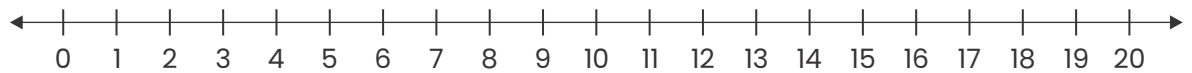
$7 + 1 = \boxed{}$


$4 + 4 = \boxed{}$


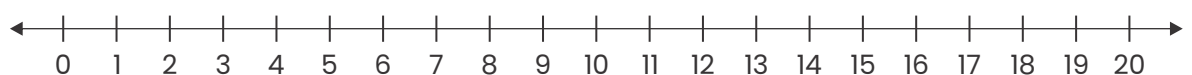
Q3. Add the following.



$3 + 6 = \boxed{}$
 $8 + 5 = \boxed{}$
 $11 + 8 = \boxed{}$



$5 + 9 = \boxed{}$
 $9 + 6 = \boxed{}$
 $11 + 5 = \boxed{}$



$3 + 2 = \boxed{}$
 $5 + 5 = \boxed{}$
 $4 + 3 = \boxed{}$

Q4. Solve the following.

(i) $(+10) + (+20)$

(ii) $(-15) + (-25)$

(iii) $(-20) + (-7)$

(iv) $(+5) + (-14)$

(v) $(-14) + (+8)$

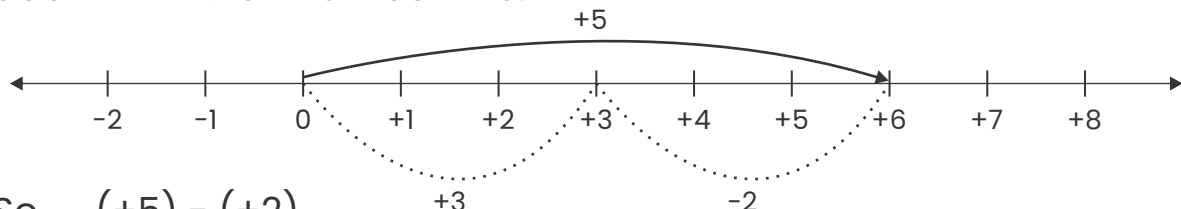
(iv) $(-30) + (+30)$

SUBTRACTION OF INTEGERS

Examples 1:

Subtract +2 from +5

Solution: First of all we change the sign of +2 and get -2 then add -2 in +5 on number line.

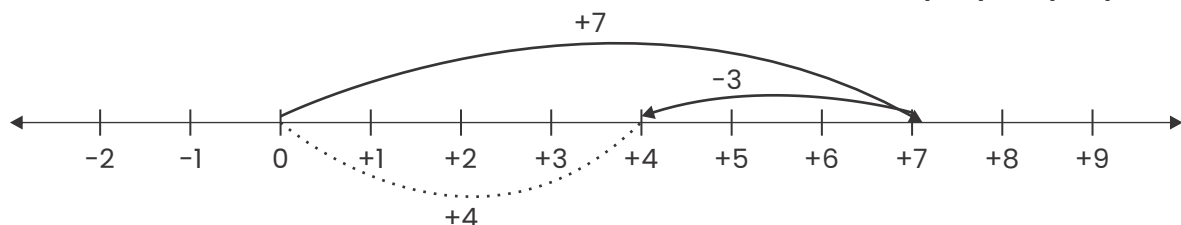


$$\begin{aligned}\text{So, } & (+5) - (+2) \\ & = (+5) + (-2) \\ & = +3\end{aligned}$$

Examples 2:

Subtract +3 from +7 using number line.

Solution: To subtract +3 from +7 means; to solve: $(+7) - (+3)$



$$\text{Here, } (+7) - (+3) = +4$$

Examples 3:

Subtract:

(i) +2 from +6 **(ii)** -3 from +8

Solution:

i) +2 from +6

Subtract +2 from +6 means to find difference between (+6) and (+2)

$$\begin{aligned}\text{Difference} & = (+6) - (+2) \\ & = (+6) + (-2) \\ & = + (6 - 2) \\ & = + 4\end{aligned}$$

∴ The sign of the answer will be positive.

(changing sign and adding)

$$|+6| = 6$$

$$|-2| = 2$$

(ii) -3 from +8

Subtract -3 from +8 means to find difference between (+8) and (-3)

$$\begin{aligned}\text{Difference} &= (+8) - (-3) \\ &= (+8) + (+3) \\ &= + (8 + 3) \\ &= + 11\end{aligned}$$

(changing sign and adding)

$$\begin{aligned}\because | +8 | &= 8 \\ | +3 | &= 3\end{aligned}$$

Note: We can solve $(+6) + (+2)$ by adding inversely as under:
 $(+6) + (+2) = (+6) - (-2) = (+6) + (+2) = + (6 + 2) = +8$

Examples: Solve:

(i) $(-8) - (-5)$ **(ii)** $(+7) - (+10)$

Solution:

i) $(-8) - (-5)$

$$\begin{aligned}&= (-8) - (-5) \\ &= (-8) + (+5) \\ &= - (8 - 5) \\ &= - 3\end{aligned}$$

$$\because 8 > 5$$

$\therefore$ The sign will be negative.

(changing sign and adding)

$$\begin{aligned}| -8 | &= 8 \\ | +5 | &= 5\end{aligned}$$

ii) $(+7) - (+10)$

$$\begin{aligned}&= (+7) - (+10) = (+7) + (-10) = - (10 - 7) \\ &= - 3\end{aligned}$$

$$\because 10 > 7$$

$\therefore$ The sign will be negative.

(changing sign and adding)

$$\begin{aligned}| -10 | &= 10 \\ | +7 | &= 7\end{aligned}$$

EXERCISE 4.3

Q1. Find the difference using number line.

(i) $(+6) - (+4)$

(ii) $(-8) - (-3)$

(iii) $(-9) - (-3)$

(iv) $(-9) - (+5)$

(v) $(+9) - (-2)$

(vi) $(-7) - (+4)$

Q2. Subtract.

(i) $+6$ from -10

(ii) -10 from -20

(iii) -8 from -7

(iv) 15 from $+20$

(v) -6 from $+14$

(vi) -9 from -6

Q3. Solve the following.

(i) $(+25) - (-15)$

(ii) $(-30) - (-25)$

(iii) $(-11) - (+5)$

(iv) $(+40) - (+30)$

(v) $(-16) - (-18)$

(vi) $(+15) - (-20)$

(vii) $(-29) - (+17)$

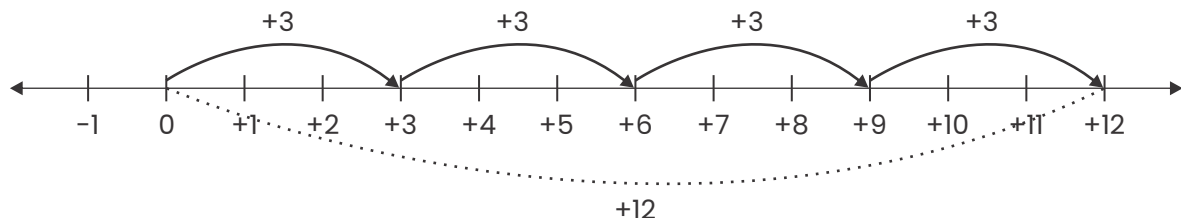
(viii) $(-20) + (-30)$

(ix) $(-27) + (+17)$

MULTIPLICATION OF INTEGERS

For example multiplying $+3$ by $+4$ means adding $+3$ four times i.e. $(+3) \times (+4) = (+3) + (+3) + (+3) + (+3)$

Representation on number line



i.e. $(+3) \times (+4) = +12$

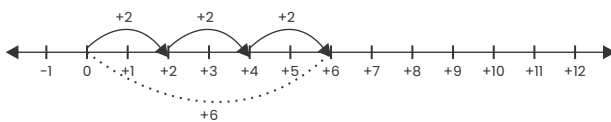
Rule 1:

The product of two integers of like signs is a positive integer.

Consider the following examples of multiplication using number line.

(i) $(+2) \times (+3)$

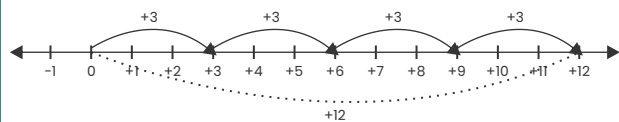
Multiplying $+2$ with $+3$ means adding $+2$ three times in same direction of $+2$.



Here $(+2) \times (+3) = +6$

(ii) $(-3) \times (-4)$

Multiplying -3 with 4 means adding -3 four times in opposite direction of -3 .



Here $(-3) \times (-4) = +12$

From these two examples we observe that.

Rule 2:

The product of two integers of unlike signs is a negative integer.

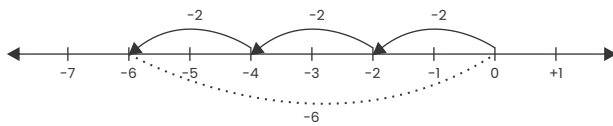
Consider the following examples of multiplication using number line.

(i) $(-2) \times (+3)$

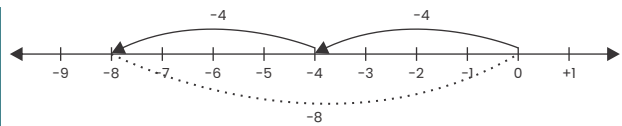
Multiplying -2 with $+3$ means adding -2 three times in same direction of -2 .

(ii) $(+4) \times (-2)$

Multiplying $+4$ with -2 means adding $+4$ two times in opposite direction of $+4$.



So, $(-2) \times (+3) = -6$



So, $(+4) \times (-2) = -8$

DIVISION OF INTEGERS

Recognize that division is the inverse process of multiplication:

i.e. $\frac{20}{5} = \boxed{} \rightarrow 20 = \boxed{} \times 5$

↓

Same number Guess it?

This number is 4.

because $\frac{20}{5} = \boxed{4} \rightarrow 20 = \boxed{4} \times 5$

Recognize that on dividing one integer by another if both the integers have like signs the quotient is positive.

Remember that

Dividing a number with any other number is in fact the product of given number with reciprocal of the divisor.

Rule 1: For integers of like signs, the quotient is positive.

For example:

(i) $20 \div (+2) = 20 \times \left(+\frac{1}{2}\right)$

$= \frac{20}{2}$

$= 10$

So, $20 \div (+2) = +10$

Rule 2: For integers of unlike signs, the quotient is negative.

Let us consider the following examples of division.

(i) $(+8) \div (-2)$

$$= (+8) \times \left(-\frac{1}{2}\right)$$

$$= -\frac{8}{2}$$

$$= -4$$

So,

$$(+8) \div (-2) = -4$$

(ii) $(-15) \div (+3)$

$$= (-15) \times \left(+\frac{1}{3}\right)$$

$$= -\frac{15}{3}$$

$$= -5$$

So,

$$(-15) \div (+3) = -5$$

From these two examples we observe that the Quotient of unlike signs is negative.

Q1. Find the product.

(i) $(+15) \times (-4)$

(ii) $(+20) \times (+17)$

(iii) $(-16) \times (-25)$

(iv) $(-36) \times (+12)$

(v) $(+35) \times (-14)$

(vi) $(-43) \times (-16)$

Q2. Find the quotient.

(i) $(+25) \div (-5)$

(ii) $(-35) \div (-7)$

(iii) $(+100) \div (+4)$

(iv) $(-252) \div (+4)$

(v) $(-234) \div (+3)$

(vi) $(-126) \div (-6)$

Q3. Write true or false.

(i) $+5 \div 0 = 0$

(iv) $0 \div +6 = 0$

(ii) $+9 \times 0 = +9$

(v) $+6 \times 0 = 0$

(iii) $(-10) \div (-10) = -1$

(vi) $(-18) \div (-3) = (+6)$